2.6.1 Division Algorithm

Given two polynomials $f(x)$ and $g(x)$, where $g(x)$ is not the zero polynomial, there exist two polynomials $q(x)$ and $r(x)$ such that $f(x) = q(x)g(x)+r(x)$ where degree of $r(x) <$ degree of $g(x)$. Here, $q(x)$ is called the quotient polynomial, and $r(x)$ is called the remainder polynomial. If $r(x)$ is the zero polynomial, then $q(x)$, $g(x)$ are factors of $f(x)$ and $f(x) = q(x)g(x)$.

These terminologies are similar to terminologies used in division done with integers.

If $g(x) = x - a$, then the remainder $r(x)$ should have degree zero and hence $r(x)$ is a constant. To determine the constant, write $f(x) = (x - a)q(x) + c$. Substituting $x = a$ we get $c = f(a)$.

Remainder Theorem

If a polynomial $f(x)$ is divided by $x-a$, then the remainder is $f(a)$. Thus the remainder $c = f(a) = 0$ if and only if $x - a$ is a factor for $f(x)$.

Definition 2.1

A real number a is said to be a ***zero of the polynomial*** $f(x)$ if $f(a) = 0$. If $x = a$ is a zero of $f(x)$, then $x - a$ is a ***factor*** for $f(x)$.

In general, if we can express $f(x)$ as $f(x) = (x - a)^k.g(x)$ where $g(a) \neq 0$, then the value of k, which depends on a, cannot exceed the degree of $f(x)$. The value k is called the ***multiplicity*** of the zero a.



(i) A polynomial function of degree n can have at most n distinct real zeros. It is also possible that a polynomial function like $P(x) = x^2 + 1$ has no real zeros at all.

(ii) Suppose that $P(x)$ is a polynomial function having rational coefficients. If $a + b\sqrt{p}$ where $a, b \in \mathbb{Q}$, p a prime, is a zero of $P(x)$, then its ***conjugate*** $a - b\sqrt{p}$ is also a zero.

Two important problems relating to polynomials are

(i) Finding zeros of a given polynomial function; and hence factoring the polynomial into linear factors and

(ii) Constructing polynomials with the given zeros and/or satisfying some additional conditions.

To address the problem of finding zeros of a polynomial function, some well known algebraic identities are useful. What is an identity?

An equation is said to be an ***identity*** if that equation remains valid for all values in its domain. An equation is called ***conditional equation*** if it is true only for some (not all) of values in its domain. Let us recall the following identities.

2.6.2 Important Identities

For all $x, a, b \in \mathbb{R}$ we have

1. $(x + a)^3 = (x + a)^2(x + a) = x^3 + 3x^2a + 3xa^2 + a^3 = x^3 + 3xa(x + a) + a^3$
2. $(x - b)^3 = x^3 - 3x^2b + 3xb^2 - b^3 = x^3 - 3xb(x - b) - b^3$ (taking $a = -b$ in (1))
3. $x^3 + a^3 = (x + a)(x^2 - xa + a^2)$
4. $x^3 - b^3 = (x - b)(x^2 + xb + b^2)$ taking $a = -b$ in (3)
5. $x^n - a^n = (x - a)(x^{n-1} + x^{n-2}a + \cdots + x^{n-k-1}a^k + \cdots + a^{n-1}),\ n \in \mathbb{N}$
6. $x^n + b^n = (x + b)(x^{n-1} - x^{n-2}b + \cdots + x^{n-k-1}(-b)^k + \cdots + (-b)^{n-1}),\ n \in \mathbb{N}$